

Threat Hunting Task: Snort and Splunk

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1 Introduction and Goals

The main goal of this task was to learn how to detect hidden cyber attacks on a network. To do this, we focused on "Proactive Threat Hunting."

Usually, security tools like firewalls act like standard alarms—they only go off if they recognize a known attack. However, smart attackers move slowly and quietly to avoid triggering these alarms. Threat hunting is the process of actively searching our network logs to find these hidden attackers.

To complete this task, we used two main tools:

- **Snort:** A network sensor that listens to traffic and creates log files of any suspicious activity.
- **Splunk:** A dashboard tool that reads those log files, makes them searchable, and helps us find mathematical anomalies (unusual spikes in traffic).

2 What We Did

2.1 Step 1: Starting Snort

First, I started Snort in listening mode. I set it to run in the background (daemon mode) on my local network interface and told it to save any alerts to a log file called `alert_fast.txt`.

```
kuchbbhi@MR-U-VN:~$ sudo snort -c /usr/local/etc/snort/snort.lua -R /usr/local/etc/snort/rules/local.rules -i lo -A alert_fast -l /var/log/snort -D
o")- Snort++ 3.10.0.0
-----
Loading /usr/local/etc/snort/snort.lua:
Loading snort_defaults.lua:
Finished snort_defaults.lua:
  references
  binder
  wizard
  appid
  js_norm
  active
  alerts
  decode
  host_cache
  host_tracker
  hosts
  network
  process
  search_engine
  so_proxy
  http2_inspect
  http_inspect
  ftp_data
  stream
  stream_ip
  stream_icmp
  stream_user
  stream_file
  arp_spoof
  back_orifice
  dns
  lmap
  netflow
  normalizer
  pop
  rnc_decode
```

Figure 1: Starting Snort in the terminal.

2.2 Step 2: Monitoring the Logs

To make sure Snort was actually catching traffic, I opened the log file using the `tail -f` command in the Linux terminal. This allowed me to actively watch the alerts, like ICMP pings and SSH scans, show up in real-time.

```
kuchbbhi@MR-U-VN:~$ sudo tail -f /var/log/snort/alert_fast.txt
01/06-03:35:49.365283 ** [1:1000002:1] "Potential SSH Scan Detected" ** [Priority: 0] (TCP) 192.168.161.128:58642 -> 192.168.161.131:22
01/06-03:35:50.626354 ** [1:1000002:1] "Potential SSH Scan Detected" ** [Priority: 0] (TCP) 192.168.161.128:39541 -> 192.168.161.131:22
01/06-03:35:51.081927 ** [1:1000002:1] "Potential SSH Scan Detected" ** [Priority: 0] (TCP) 192.168.161.128:33938 -> 192.168.161.131:22
01/06-03:35:53.004919 ** [1:1000002:1] "Potential SSH Scan Detected" ** [Priority: 0] (TCP) 192.168.161.128:42714 -> 192.168.161.131:22
01/06-03:35:54.145073 ** [1:1000002:1] "Potential SSH Scan Detected" ** [Priority: 0] (TCP) 192.168.161.128:62277 -> 192.168.161.131:22
01/06-03:44:04.668292 ** [1:1000001:1] "ICMP Ping Detected" ** [Priority: 0] (ICMP) fe08:13f50:af9a:ffda:da5a -> ff02::16
01/06-03:44:04.740785 ** [1:1000001:1] "ICMP Ping Detected" ** [Priority: 0] (ICMP) fe08:13f50:af9a:ffda:da5a -> ff02::16
01/06-03:44:04.740837 ** [1:1000001:1] "ICMP Ping Detected" ** [Priority: 0] (ICMP) fe08:13f50:af9a:ffda:da5a -> ff02::16
01/06-03:44:04.741076 ** [1:1000001:1] "ICMP Ping Detected" ** [Priority: 0] (ICMP) fe08:13f50:af9a:ffda:da5a -> ff02::16
01/06-03:44:04.851856 ** [1:1000001:1] "ICMP Ping Detected" ** [Priority: 0] (ICMP) fe08:13f50:af9a:ffda:da5a -> ff02::16
07/00-14:07:13.590765 ** [1:366:11] "PROTOCOL-ICMP Ping Unix" ** [Classification: Misc activity] [Priority: 3] (ICMP) 127.0.0.1 -> 127.0.0.1
07/00-14:07:13.590765 ** [1:1000001:1] "ICMP Ping Detected" ** [Priority: 0] (ICMP) 127.0.0.1 -> 127.0.0.1
07/00-14:07:13.590765 ** [1:29456:3] "PROTOCOL-ICMP Unusual Ping detected" ** [Classification: Information Leak] [Priority: 2] (ICMP) 127.0.0.1 -> 127.0.0.1
07/00-14:07:13.590778 ** [1:1000001:1] "ICMP Ping Detected" ** [Priority: 0] (ICMP) 127.0.0.1 -> 127.0.0.1
07/00-14:07:13.590778 ** [1:480:18] "PROTOCOL-ICMP Echo Reply" ** [Classification: Misc activity] [Priority: 3] (ICMP) 127.0.0.1 -> 127.0.0.1
07/00-14:07:14.597201 ** [1:366:11] "PROTOCOL-ICMP Ping Unix" ** [Classification: Misc activity] [Priority: 3] (ICMP) 127.0.0.1 -> 127.0.0.1
07/00-14:07:14.597201 ** [1:1000001:1] "ICMP Ping Detected" ** [Priority: 0] (ICMP) 127.0.0.1 -> 127.0.0.1
07/00-14:07:14.597201 ** [1:29456:3] "PROTOCOL-ICMP Unusual Ping detected" ** [Classification: Information Leak] [Priority: 2] (ICMP) 127.0.0.1 -> 127.0.0.1
07/00-14:07:14.597216 ** [1:1000001:1] "ICMP Ping Detected" ** [Priority: 0] (ICMP) 127.0.0.1 -> 127.0.0.1
07/00-14:07:14.597216 ** [1:480:18] "PROTOCOL-ICMP Echo Reply" ** [Classification: Misc activity] [Priority: 3] (ICMP) 127.0.0.1 -> 127.0.0.1
07/00-14:07:15.621007 ** [1:366:11] "PROTOCOL-ICMP Ping Unix" ** [Classification: Misc activity] [Priority: 3] (ICMP) 127.0.0.1 -> 127.0.0.1
07/00-14:07:15.621007 ** [1:1000001:1] "ICMP Ping Detected" ** [Priority: 0] (ICMP) 127.0.0.1 -> 127.0.0.1
07/00-14:07:15.621007 ** [1:29456:3] "PROTOCOL-ICMP Unusual Ping detected" ** [Classification: Information Leak] [Priority: 2] (ICMP) 127.0.0.1 -> 127.0.0.1
07/00-14:07:15.621021 ** [1:1000001:1] "ICMP Ping Detected" ** [Priority: 0] (ICMP) 127.0.0.1 -> 127.0.0.1
07/00-14:07:15.621021 ** [1:480:18] "PROTOCOL-ICMP Echo Reply" ** [Classification: Misc activity] [Priority: 3] (ICMP) 127.0.0.1 -> 127.0.0.1
07/00-14:07:15.621021 ** [1:366:11] "PROTOCOL-ICMP Ping Unix" ** [Classification: Misc activity] [Priority: 3] (ICMP) 127.0.0.1 -> 127.0.0.1
07/00-14:07:16.645028 ** [1:1000001:1] "ICMP Ping Detected" ** [Priority: 0] (ICMP) 127.0.0.1 -> 127.0.0.1
07/00-14:07:16.645028 ** [1:29456:3] "PROTOCOL-ICMP Unusual Ping detected" ** [Classification: Information Leak] [Priority: 2] (ICMP) 127.0.0.1 -> 127.0.0.1
07/00-14:07:16.645044 ** [1:1000001:1] "ICMP Ping Detected" ** [Priority: 0] (ICMP) 127.0.0.1 -> 127.0.0.1
07/00-14:07:16.645044 ** [1:480:18] "PROTOCOL-ICMP Echo Reply" ** [Classification: Misc activity] [Priority: 3] (ICMP) 127.0.0.1 -> 127.0.0.1
07/00-14:09:42.126243 ** [1:1000001:1] "ICMP Ping Detected" ** [Priority: 0] (ICMP) 127.0.0.1 -> 127.0.0.33
07/00-14:09:42.126243 ** [1:482:16] "PROTOCOL-ICMP destination unreachable port unreachable packet detected" ** [Classification: Misc activity] [Priority: 3] (ICMP) 127.0.0.1 -> 127.0.0.1
```

Figure 2: Actively monitoring Snort alerts using the tail command.

2.3 Step 3: Adding Logs to Splunk

Reading text files in the terminal is difficult, so I added the Snort log file to Splunk. Splunk monitors the file and puts the data into a dashboard where I can easily search and filter the alerts.

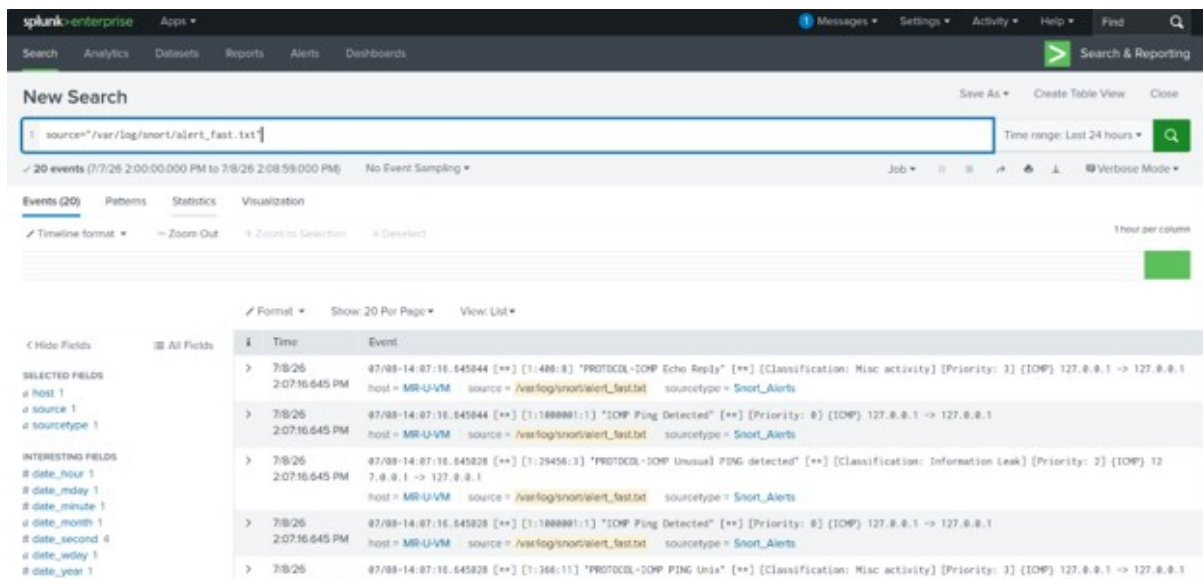


Figure 3: Monitoring the Snort log file inside Splunk.

2.4 Step 4: Threat Hunting with Splunk

To catch slow attacks that evade basic Snort rules, I wrote a custom search in Splunk. This query counts the number of events, calculates the average to find a "normal baseline," and sets a threshold. If the traffic spikes above that threshold, Splunk flags it as a "THREAT DETECTED."

```
source="/var/log/snort/alert_fast.txt"
| timechart count by Event
| streamstats avg(count) as baseline stdev(count) as standard_deviation window=5
| eval anomaly_threshold = baseline + (standard_deviation * 2)
| eval is_anomaly = if(count > anomaly_threshold, "YES - THREAT DETECTED",
  "Normal Baseline")
```



Figure 4: Splunk search query detecting the anomaly.

3 What We Can Improve

To make this project better in the future, we can implement the following upgrades:

- **Automate Startup:** Instead of typing commands to start Snort and Splunk every time the computer turns on, we can set them up as Ubuntu **systemd** services. This means both tools will start automatically in the background as soon as the machine boots up, ensuring the network is always monitored without manual effort.
- **Convert to an IPS (Active Defense):** Right now, Snort is running as a passive Intrusion Detection System (IDS), meaning it only logs the attacks. We can upgrade this to an active Intrusion Prevention System (IPS). By running Snort in "Inline Mode" and changing our rules from **alert** to **drop**, Snort will actively destroy malicious packets before they enter the network.